REASEARCH ASSIGNMENT 1

1. DATABASE

= an organised collection of data that is ~~use~~ stored and managed so it can be easily accessed and updated

* Databases are essential in the modern world because almost every system relies on data to function efficiently and accurately. They help store large amounts of information securely, reduce duplication, and enable quick decision-making in businesses and organisations

* REAL-WORLD EXAMPLES:

i) Banks use databases to store customer details, account balances, and transaction histories

ii) Social media platforms use databases to manage user profiles, posts, messages, and interactions

2. DATA WAREHOUSE

= a central system that stores large amounts of structured data collected from different sources.

* A regular database is used for daily operations, like inserting, updating, and retrieving current data quickly. A data warehouse is different because it is used to analyse past data for complex quiries rather than everyday transactions

* Companies use a data warehouse to combine data from multiple systems into one place for a complete view of their operations. It helps in decision-making by identifying patterns, tracking performance, and ~~sup~~ supporting business intelligence tools.

3. DATA LAKE

= a system that stores large amounts of raw data in its original form, without structuring it first.

* A data lake stores raw, unprocessed data ~~that is~~ and gives more flexibility, but requires processing before it is used. Data warehouse stores structured and processed data that is cleaned and ready for use.

* A business would use a data lake when it needs to store large volumes of diverse data, like logs, images, or sensor data.

4. DATA MART

= a smaller section of data warehouse designed for a specific department or business area.

* A data mart takes, from data warehouse, a small portion of data and organises it for a specific use, like sales, finance, or marketing

* It is usually used by business teams such as sales, finance, or marketing to get insights relevant to their work.

5.

STRUCTURED DATA	SEMI-STRUCTURED DATA	UNSTRUCTURED DATA
= Data that is highly organised and stored in a fixed format, usually in tables with rows and columns.	= Data that does not follow a strict table format but still has some organisation through tags or markers.	= Data that has no predefind format or structure, making it harder to store and analyse.
e.g Customer information	e.g JSON or XML files used in web applications	eg Images, videos, audio files ...

6. DATA MODELING

= The process of defining how data is structured, stored, and related within a system.

* Data modeling ensures that data is organised efficiently, reducing errors and duplication. It also makes systems easier to design, maintain, and scale.

* i) Conceptual Data Model
  ii) Logical Data Model
  iii) Physical Data Model

7. DATA MINING

= The process of analysing large sets of data to discover patterns, trends, and other useful insights.

* Businesses use data mining to make better decisions by identifying customer behaviour, predicting trends, and improving operations.

* - Online stores analyse customer purchase history to suggest products they are likely to buy.
  - Banks use data mining to detect unusual transaction to detect fraud

8. DATABASE MANAGEMENT SYSTEM (DBMS)

= A software that allows users to create, store, manage, and retrieve data in a database.

(a) MySQL - an open-source relational DBMS that stores data in tables with rows and columns
- Commonly used for web applications like websites and e-commerce platforms due to its reliability and ease of use.

(b) PostgreSQL - an advanced open-source relational DBMS known for strong performance and support for complex queries
- Ideal for applications that require complex data operations such as financial systems or analytics platforms

(c) Oracle Database - a powerful commercial DBMS designed for large-scale enterprise-level applications
- Used by large organisations for handling massive volumes of secure data like banking systems

(d) Mongo DB - a DBMS that stores data in flexible, JSON-like documents instead of tables
- Ideal for applications with rapidly changing data structures.

9. ETL (EXTRACT, TRANSFORM, LOAD)

= A process used to collect data from different sources, prepare it, and store it in a system like a data warehouse, for analysis.

- EXTRACT: Collection of data from various sources
- TRANSFORM: Extracted data is cleaned, organised, and converted into suitable format.
- LOAD: This final step involves storing the transformed data into a targeted system like a data warehouse

* ETL ensures data quality by cleaning and standardising information before it is used.

10. (a) CSV (COMMA - SEPARATED VALUES)

= A simple text format where data is stored in rows and columns separated by commas

* Best for simple data exchange and small datasets because it is easy and widely supported.

(b) JSON (JavaScript Object Notation)

= A flexible, semi-structured format that stores data as key-value pairs.

* Ideal for web applications and APIs where data structure may vary or be nested.

(c) PARQUET -

= A columnar storage format optimised for big data processing and analytics

* Used in big data environments when you need fast query performances and efficient storage, especially for large datasets.

(d) AVRO

= A row-based format that supports schema evolution, meaning the structure can change over time.

* Useful in data streaming and systems where data structure may evolve, such as pipelines using distributed systems.

ARTIFICIAL INTELLIGENCE (AI)

= A field of computer sciend that focuses on creating systems that can perform tasks that normally require human intelligence

- NARROW AI - designed to perform a specific task
  e.g ChatGPT
- GENERAL AI - a theoretical type of AI that would have human-like intelligence and the ability to understand, learn, and apply knowledge across a wide range of tasks

e.g A fully autonomous system that can independently perform any intellectual task a human can do, such as reasoning, learning new skills, and adapting to completely new environments

12. MACHINE LEARNING (ML)

= A branch of AI where computers learn patterns from data and improve their performance over time without being explicitly programmed for ~~every task~~

* In traditional programming, a developer writes fixed rules and instructions for the computer to follow. In machine learning, the system is given data and learns the rules and patterns on its own to make predictions or decisions

TYPES:

- SUPERVISED: This model is trained using labeled data.
- UNSUPERVISED: This model is given unlabeled data and must find patterns or groupings on its own.
- REINFORCEMENT: The model learns by interacting with an environment and receiving rewards or penalties for its actions

13. DEEP LEARNING

= subset of Machine Learning that uses complex algorithms called neural networks to learn patterns from large amounts of data

- Machine Learning is a subset of AI that focuses on learning from data, and Deep Learning is a further subset of ML that uses multi-layered neural networks to model complex patterns

NEURAL NETWORKS: Computational models inspired by the human brain, made up of layers of interconnected nodes/neurons

- Data enters the input layer, passes through one or more hidden layers, and produces output in the final layer.

= a branch of AI that enables computers to understand, interpret, and generate human language.

REAL-WORLD APPLICATIONS:

~~• VIRTUAL ASSISTANTS:~~
- Virtual assistants
- Text messaging and autocorrect
- Chatbots and customer support

15. LARGE LANGUAGE MODEL (LLM)

= a type of AI model trained on vast amounts of text data to understand, generate, and respond to human language

* An LLM is trained on a wide range of language data and can perform many different language-related tasks without being retrained for each one, unlike SML which is only for a specific task.

~~• GPT-4~~: Used for tasks like conversation, content creation, coding, and tutoring

• BERT: Used to improve search engines and understand the context of words in queries

16 GENERATIVE AI

= a type of AI that can create new content such as text, images, audio, or code based on patterns it has learned from existing data.

- Generative AI models are trained on large datasets to learn patterns, structure, and relationships in the data. When given a prompt, the model uses those learned patterns to generate new content relevant and coherent

- ChatGPT: Generates human-like text for conversations, writing, coding, and answering questions

- DALL-E: Creates images from text descriptions, allowing users to generate artwork or visuals from prompts

GitHub Copilot: Assists by generating code suggestions and completing functions while they are programming

17. TRAINING DATASETS

= a collection of data used to teach an AI or machine learning model how to recognise patterns and make decisions.

- High-quality data ensures the model learns accurate and meaningful patterns
- If the training data is incorrect or incomplete, the model may learn wrong patterns, leading to poor predictions and unreliable decisions

18 COMPUTER VISION

= a field of AI that enables machines to interpret and understand visual information from images & videos

- Images are made up of pixels, each containing numerical values representing color and intensity

Computer vision systems use algorithms and deep learning models to detect patterns, edges, shapes, and objects, then interpret what those patterns represent.

REAL-WORLD CASE:

- Facial recognition
- Self-driving cars

19. API (APPLICATION PROGRAMMING INTERFACE)

= a set of rules that allows different software systems to communicate and share data with each other.

An API is like a waiter in a resturant. You tell it what you want, and it takes your request to the kitchen, and brings back your order.

- APIs allow different systems to connect and wo
  together efficiently.

20. API key

= a unique code used to identify and authenticate a user or application when it makes a request to an API

- Purpose of an API is to help control access to an API
- API keys should be kept private and never shared in public places like websites
- If someone gains access to the API key, they can misuse the API as if they were the legitimate user.

21. REST API

= a type of API that follows a set of rules for how systems communicate over web using standard HTTP meth

MAIN HTTP METHODS:

- GET: Used to retrieve data from a server
- POST: Used to send data to the server
- PUT: Used to update an existing resource to the serv
- DELETE: Used to remove a resource from the server

22. JSON (JavaScript Object Notation)

= a lightweight data format used to store and exchange data between systems

- JSON is commonly used because it is simple, fast, and language-dependant, which makes it ideal for communication

</> JSON

```
{
  "name": "John",
  "age": 25,
  "email": "john@example.com",
  "isStudent": false
}
```

23 PROMPT ENGINEERING

= skill of designing and refining inputs given to an AI model to get more accurate, useful, and relevant outputs
. AI tools respond based on how a question is asked. That's why its important.

EXAMPLES:
- Tell me about AI
- Help me with my assignment.

24. MAP FUNCTION

= a programming tool used to apply a specific operation to every item in a collection and return a new collection with the results
. MAP function is commonly used to clean, transform, or modify data efficiently

```
numbers = [1, 2, 3, 4, 5]
squared_numbers = list(map(lambda x: x*x, numbers))
print(squared_numbers)
```

output

[1, 4, 9, 16, 25]

25. CLOUD COMPUTING

= is the delivery of computing services over the internet instead of using local computers or physical infrastructure

- **IaaS**: Provides basic computing infrastructure such as virtual machines, storage, and networks.
  e.g Amazon Web Services.
- **PaaS**: Provides a platform ~~full~~ that allows ~~development~~ developers to build, test, and deploy applications without managing the underlying infrastructures
  e.g Google Cloud Platform.
- **SaaS**: Delivers fully built software applications over the internet that users can access through a browser
  e.g Microsoft 365

26. VERSION CONTROL SYSTEM (VCS)

= Allows multiple people to collaborate on a project, keep a history of changes, and revert to earlier versions if needed

GIT - a distributed version control system that tracks changes in code and files efficiently.

- It is widely used by developers and data analysts because it supports collaboration, maintains a full history of changes, and works well for managing projects ~~of~~ any size. Makes teamwork easier. Helps in tracking errors

COMMIT: A saved snapshot of changes made to files in a project.

BRANCH: A separate line of development within a project.

27. JUPYTER NOTEBOOK

= an interactive web-based tool used to write and run code, view outputs and add explanations in one document.

- Jupyter Notebook is commonly used with Python.

WHY IT IS POPULAR:

- Users can run code step by step and see results immediately.
- It easily supports graphs, charts, and plots.
- Users can document their thinking process alongside the code

28 PYTHON

= a high-level, general-purpose programming language known for its simple and reliable syntax

- Python is easy to learn and has a large ecosystem of libraries

COMMON PYTHON LIBRARIES

- Numpy: Used for numerical operations
- Pandas: Used for data manipulation
- Matplotlib: Used for creating graphs, charts, and data visualisation
- Scikit-learn: Used for building and training machine learning models

29. DATA PRIVACY

= refers to the protection of personal information and how it is collected, stored and used by organisations

- Data privacy is important because personal data can be sensitive, and if misused it can lead to identity theft, fraud, or loss of trust.

GDPR (GENERAL DATA PROTECTION REGULATION)

= Data protection law introduced by the European Union in 2018.

- It applies to all countries in the European Union and European Economic Area.

30. DATA ETHICS

= refers to the moral principles and guidelines that govern how data is collected, used, shared, and analysed.

EXAMPLES;

- UNAUTHARISED DATA COLLECTION

When companies collect personal data without user consent.
- This is problematic because it violates privacy rights.

- BIASED AI DECISION-MAKING

When an AI system is trained on biased data and produces unfair outcomes.
- This is problematic because it leads to discrimination and unfair treatment of certain groups

- DATA ~~MISU~~ MISUSE OR SELLING PERSONAL INFORMATION

When organisations sell or share user data with third parties without clear consent.
- This is problematic because it can expose sensitive information and break trust between users and organisations

32 BIAS IN AI

= refers to systematic errors in an AI system that cause it to produce unfair or inaccurate results for certain groups of people

- An AI model learns from historical data, If that data reflects existing human prejudices or is not representative of all groups, the model will learn and repeat those mistakes or patterns - This results in decisions that are not neutral or fair

- Unfair discrimination against certain groups (race, gender, age)
- Reduced trust in AI systems
- Legal and ethical issues for companies

31. DATA GOVERNANCE

= a set of policies, processes, and standards that ensure data is managed properly

KEY COMPONENTS

- DATA QUALITY: Ensures data is accurate
- DATA SECURITY AND PRIVACY: Protects data from unauthorised access
- DATA OWNERSHIP AND ACCOUNTABILITY: Clearly defines who is responsible for managing and maintaining data.
- DATA POLICIES AND STANDARDS: Establishes rules for how data should be collected and used.
- ~~DATA LIFECYCLE~~
- DATA LIFECYCLE MANAGEMENT: Manages data from creation to deletion.

ROLE	MAIN FOCUS	KEY SKILLS	COMMON TOOLS
Data analyst	Analysing historical data to find insights and support business decisions	Data cleaning, visualisation, reporting, SQL, basic statistics	Excel, SQL, Power BI, Tableau, Python
Data scientist	Building predictive models and advanced analytical skills	Machine learning, statistics, data modeling, programming,	Python, R, Scikit-learn, TensorFlow
Data engineer	Designing and maintaining data pipelines and infrastructure	Data pipelines, ETL processes, database systems, cloud platforms	Apache Spark, Hadoop, SQL, cloud services
AI engineer	Developing AI systems and deploying machine learning models into production	Deep learning, neural networks, model deployment.	TensorFlow, PyTorch, Python, cloud AI platforms

3.4 BUSINESS INTELLIGENCE (BI)

= refers to the processes, technologies, and tools used to collect, analyse, and visualise business data in order to support better decision-making

POWER BI: Microsoft tool used to create interactive dashboards and reports

TABLEAU: Allows users to build interactive graphs, charts and dashboards

- BI tools combine data from multiple sources and present it in a clear, visual format. This allows decision making to be easier. Businesses can make faster, more accurate, and evidence-based decisions

35. AI

It is surprising because it covers a wide range of abilities that normally require human intelligence. I am eager to learn what I can personally do with it in my life

36. DATA MODELING

= The most difficult concept to understand - for me, is data modeling. I had to use simple online explanations and beginner tutorials on data modeling, together with visual diagrams and lecture notes

37. - In everyday life, AI will continue to influence how I interact with technology like YouTube, Netflix.
- AI assistants and chatbots can also help me with studying, answering questions, and organising tasks

38.

1. Open AI (2024)
2. Microsoft (2024)
3. Google (2024)
4. Amazon Web Services (2024)
5. IBM (2024)
6. Tableau Software (2024)
7. Scikit-learn developers (2024)
8. Geeks for Geeks (2024)
9. Lecture Notes